

DEVOPS SESSION TWO





PRESENTER INFORMATION

Russell A. Miller, BSME, OOTE, PST, LSS
Green Belt, CCBA

Consultant - Product Development Leader -
Agile Coach - Scrum.org Professional
Trainer - Mechanical & Aerospace Design
Engineer - Analyst

[linkedin.com/in/russellamiller](https://www.linkedin.com/in/russellamiller)



AGENDA

Completing our introduction to DevOps with practical insights.

- **Recap of Session 1**
- **DevOps History and Evolution**
- **Major DevOps Tools and Platforms**
- **Infrastructure as Code Overview**
- **Collaboration and Communication**
- **Metrics for Success**
- **Group Exercise: DevOps Expectations**

RECAP OF SESSION 1



Key Takeaways from Last Session

DevOps Definition and Culture

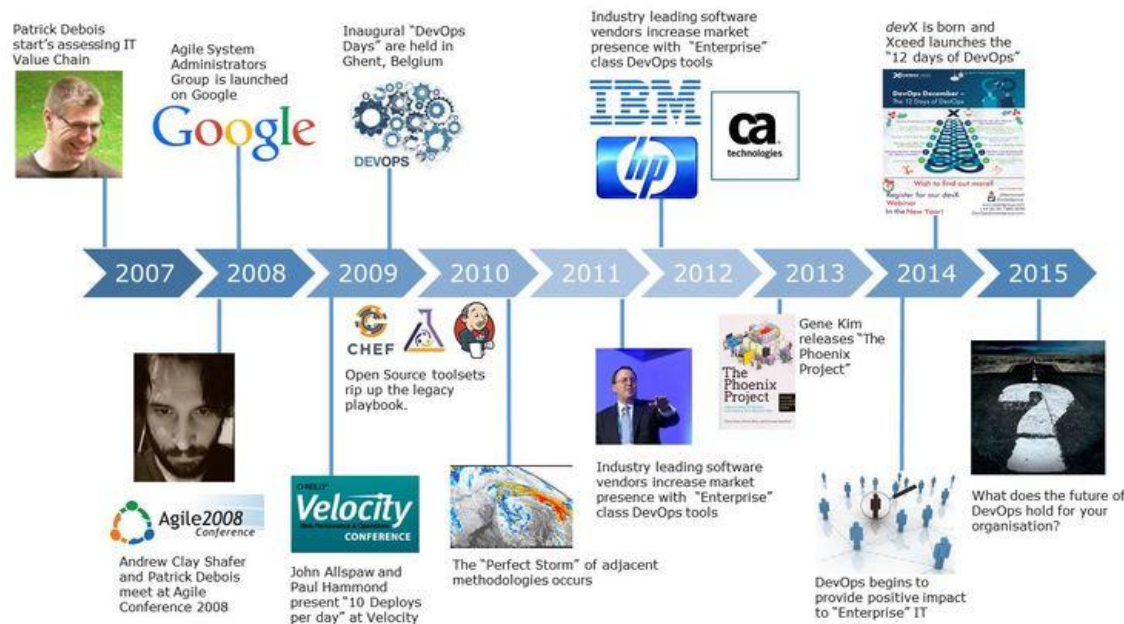
Three Ways: Flow, Feedback, Learning

Initial Benefits Explored



❖ DEVOPS HISTORY AND EVOLUTION

DevOps Timeline



- Origins of DevOps
- Evolution Over Time
- Industry Adoption Trends

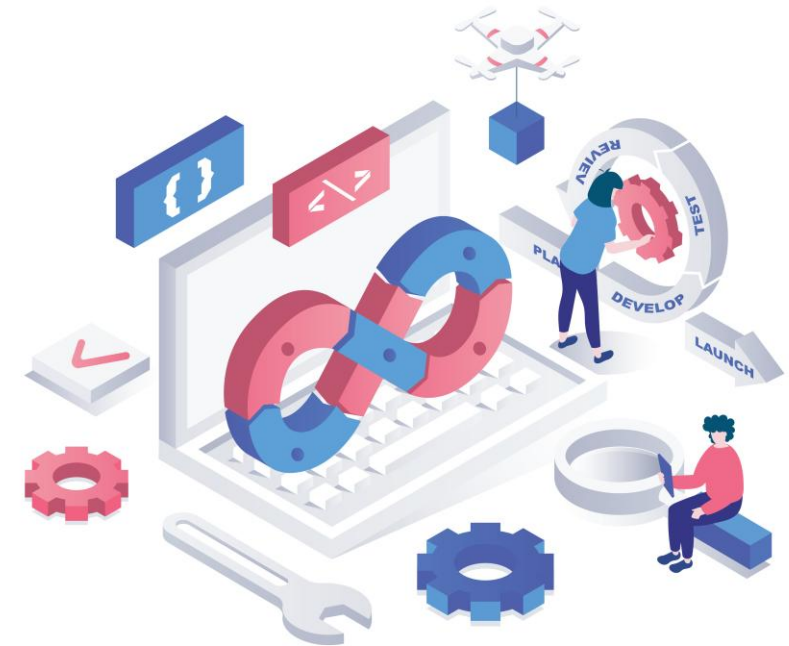
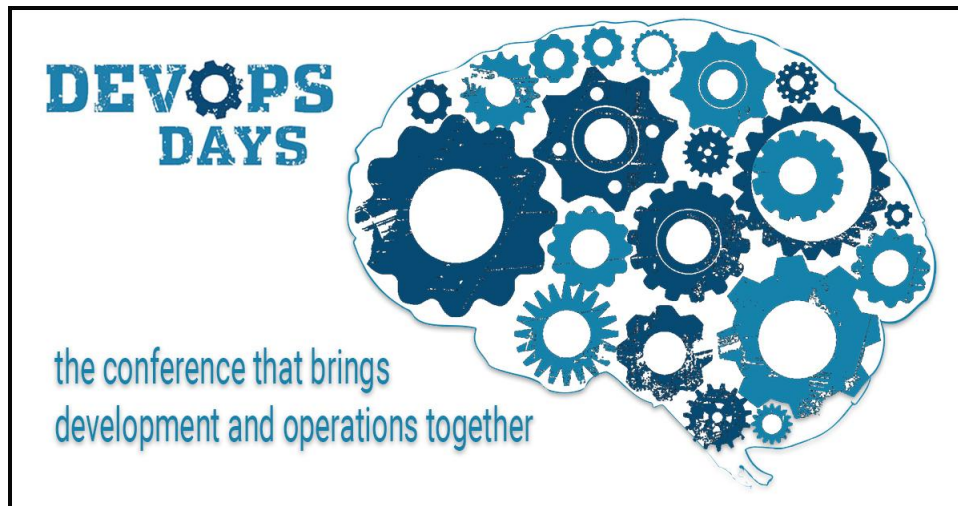
ORIGINS OF DEVOPS

The Birth of a Movement

Agile and IT Pain Points

Patrick Debois and DevOpsDays

Early Influencers



EVOLUTION OVER TIME

Growth of DevOps Practices

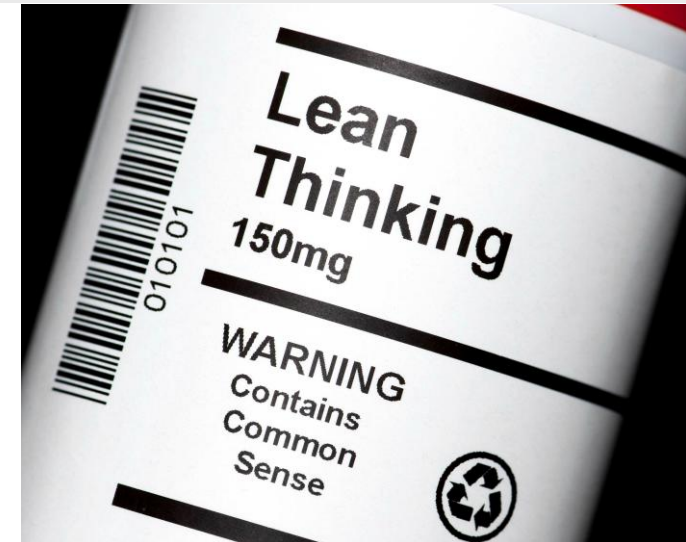
Integration with Lean and Agile

Rise of Automation

Cultural Shift Emphasis



Automation



INDUSTRY ADOPTION TRENDS

DevOps Goes Mainstream

Tech Giants Lead the Way

Cross-Industry Spread

Current State

NETFLIX



❖ MAJOR DEVOPS TOOLS AND PLATFORMS



Automation Tools
Collaboration Platforms
Monitoring Solutions



AUTOMATION TOOLS

Streamlining Processes

Jenkins for CI/CD/CD

Docker for Containers

Ansible for IaC



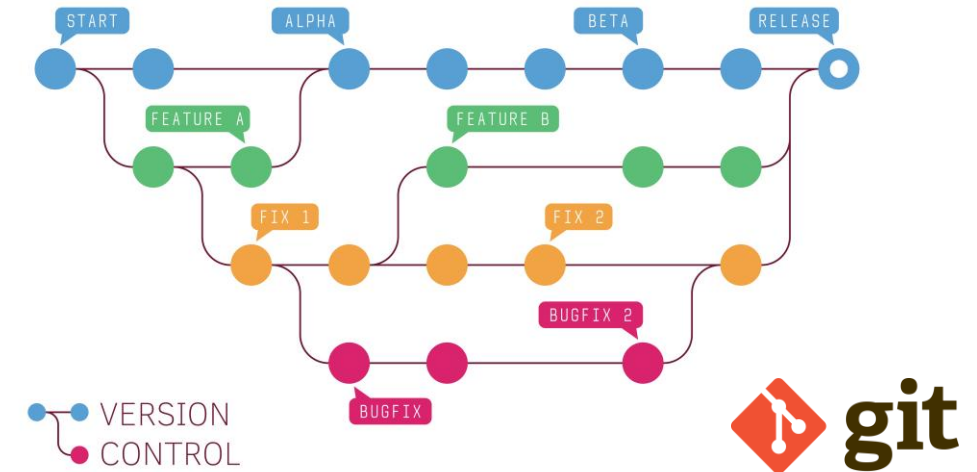
COLLABORATION PLATFORMS

Enhancing Teamwork

Slack for Communication

Git for Version Control

Jira for Task Management



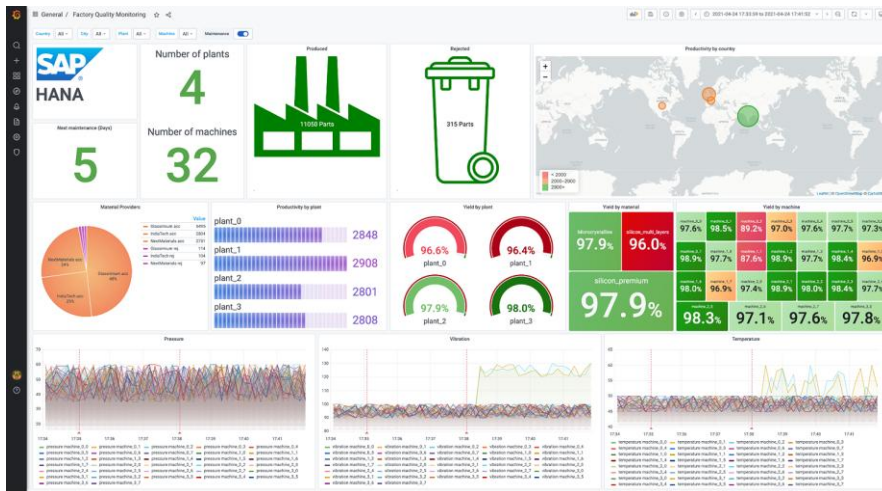
MONITORING SOLUTIONS

Ensuring Visibility

Prometheus for Metrics

Nagios for Alerts

Grafana for Dashboards



An illustration of a person with dark hair and glasses, wearing a red sweater and dark pants, sitting on a blue block and working on a laptop at a blue desk. The person is enclosed within a large, light gray cloud shape. Surrounding the person are several floating icons: a white circle with a cross, a red Wi-Fi symbol, a red wrench and screwdriver crossed, a red arrow pointing up with a gear, and a gray gear. To the right of the person, there are two blue server racks and a gray gear. Below the server racks, there are three red horizontal bars and a red target symbol. The entire scene is set against a white background.

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CONFIGURATION

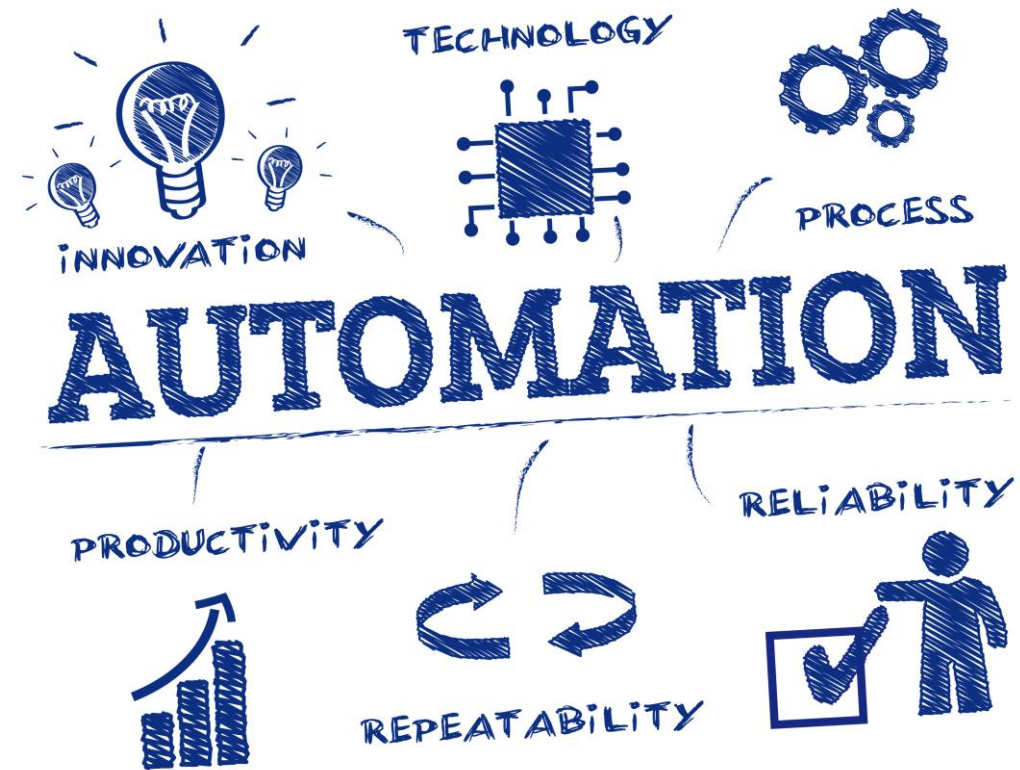
WHAT IS IAC?

Defining Infrastructure as Code

Code-Driven Infrastructure

Declarative Approach

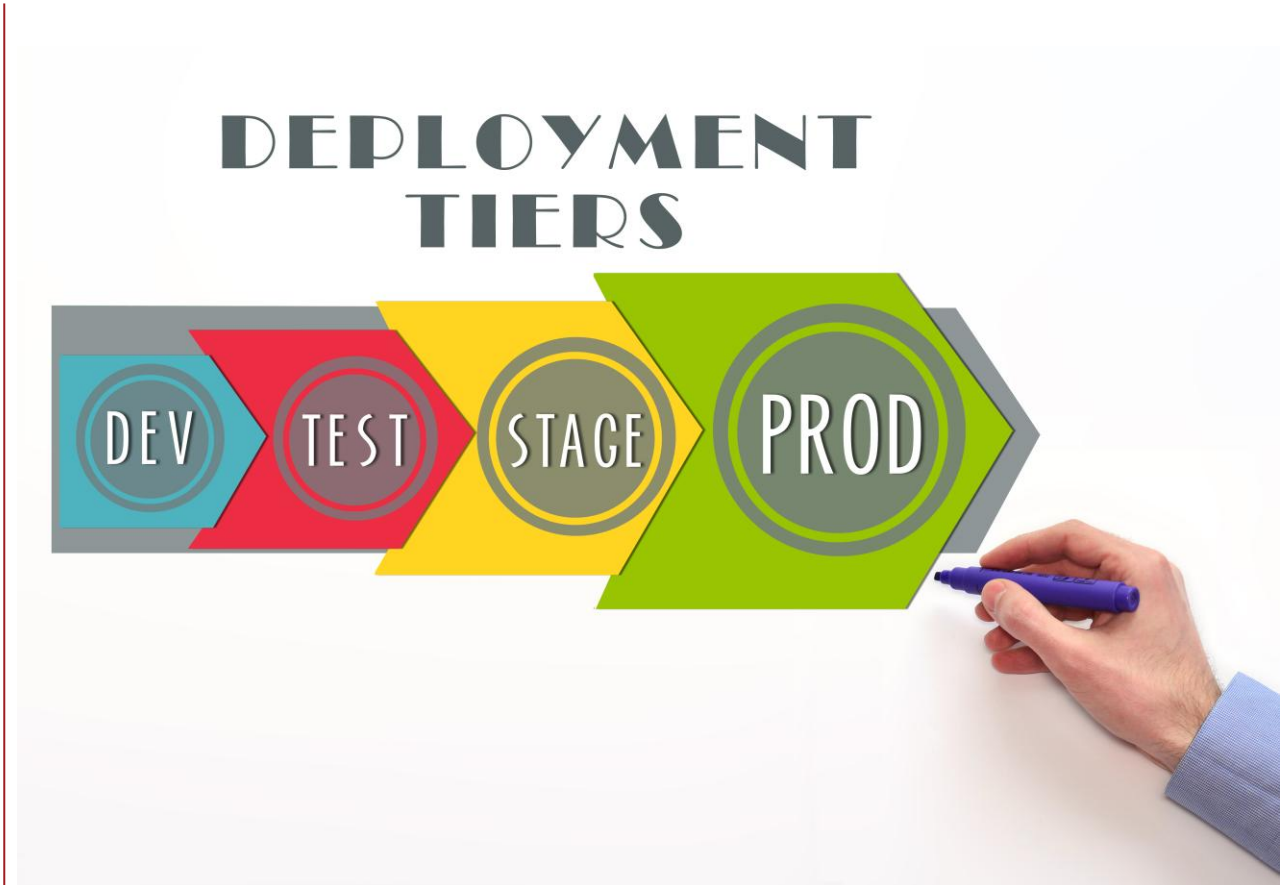
Automation Focus



WHY IT MATTERS

Benefits of IaC

- Consistency Across Environments
- Faster Provisioning
- Scalability Support



BASIC EXAMPLES



ANSIBLE

laC in Action

- Terraform Example
- Ansible Playbook
- CloudFormation Snippet



AWS CloudFormation

BASIC TERRAFORM CONFIG

```
resource "aws_instance" "Web" {  
  ami          = "ami-07013dd48140efd73" # Replace with your desired AMI ID  
  instance_type = "t3.micro"  
  tags = {  
    Name = "Ubuntu Server"  
  }  
  # subnet_id    = "vpc-0bf039e4e823f5223" # Replace with your subnet ID  
  key_name       = "AWS_Ubuntu_RSA"  
  # security_groups = ["Everybody All In"] # Ensure the security group exists with  
  this name  
}
```



BASIC ANSIBLE PLAYBOOK

- hosts: all

become: yes

tasks:

- name: Task - 1 Update APT package manager repositories cache

become: true

apt:

update_cache: yes

- name: Task -2 Install Java using Ansible

become: yes

apt:

name: "{{ packages }}"

state: present

vars:

packages:

- openjdk-11-jdk



❖ COLLABORATION AND COMMUNICATION



- Breaking Down Silos
- Tools for Collaboration
- Cultural Shifts



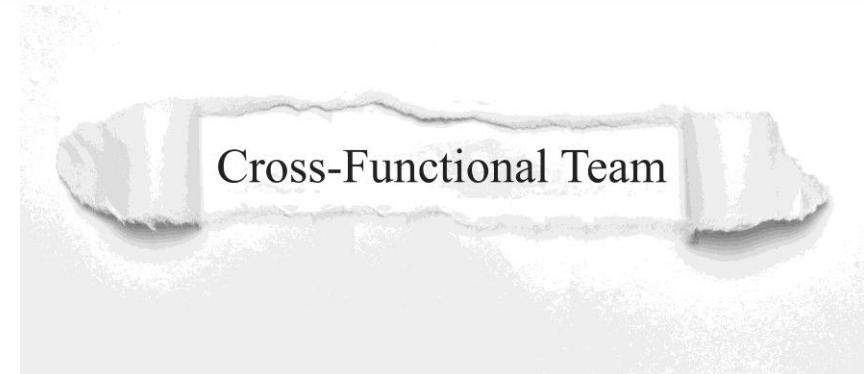
BREAKING DOWN SILOS

Fostering Cross-Functional Teams

Shared Goals

Cross-Team Metrics

Joint Decision-Making



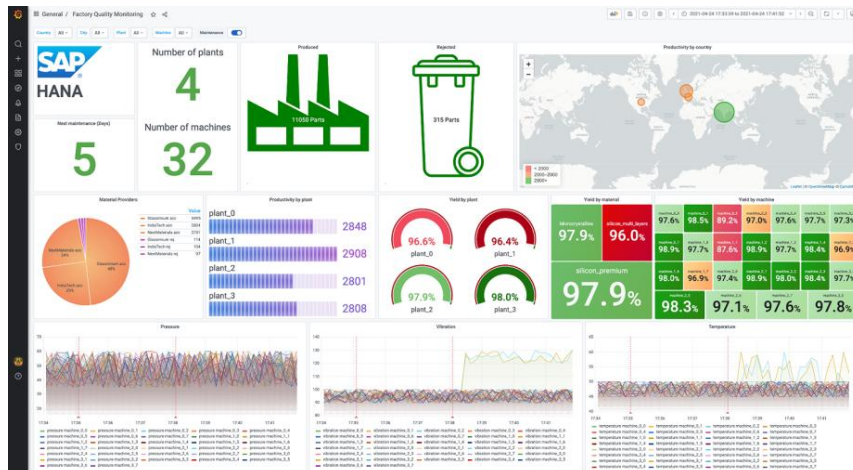
TOOLS FOR COLLABORATION

Enhancing Team Interaction

Slack, Teams

Kanban Boards

Shared Dashboards



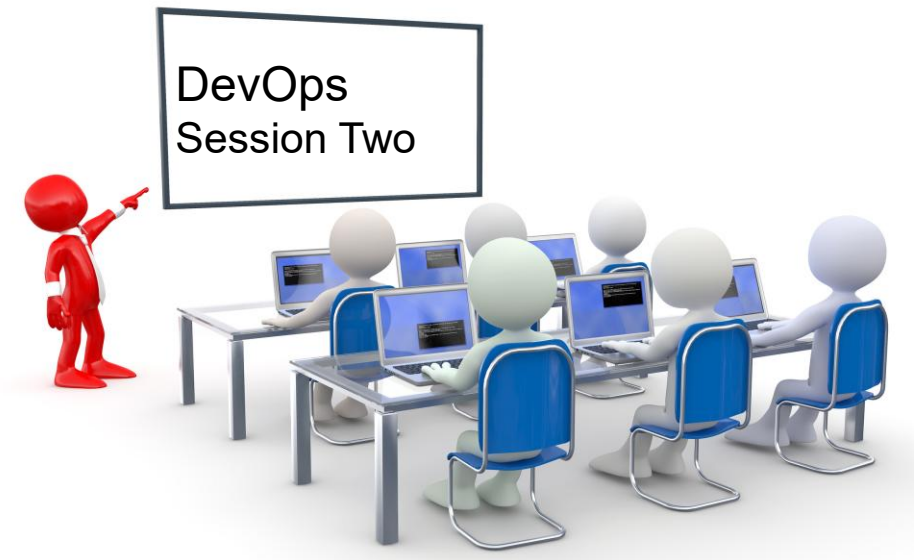
CULTURAL SHIFTS

Promoting DevOps Culture

Continuous Learning

Blameless Culture

Innovation Encouragement



❖ **METRICS FOR SUCCESS**

- **Key Performance Indicators**
- **Aligning Metrics with Business**
- **Measuring Impact**



KPI

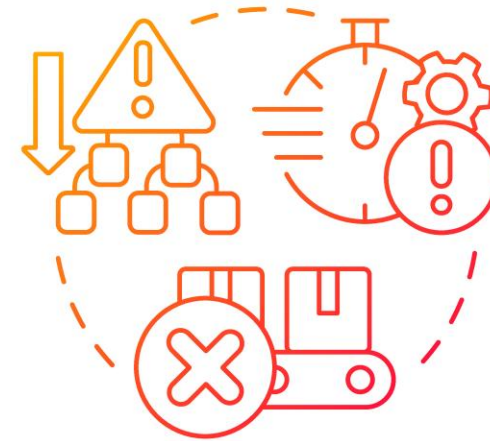
KEY PERFORMANCE INDICATORS

Understanding DevOps Metrics

Lead Time

Deployment Frequency

Change Failure Rate



**POOR BUILD QUALITY
& RELEASE SPEED**

ALIGNING METRICS WITH BUSINESS

Tying DevOps to Business Goals

Efficiency

Customer Satisfaction

Risk Reduction



MEASURING IMPACT

Quantifying DevOps Benefits

Before and After Comparisons

Benchmarking

Continuous Improvement



BEFORE

AFTER

**CONTINUOUS
IMPROVEMENT**



GROUP EXERCISE: DEVOPS EXPECTATIONS

Form Groups

Discuss Expectations

Present Insights



QUIZ QUESTION 1

What sparked the DevOps movement?

- A. New hardware advances
- B. Agile adoption challenges
- C. Regulatory changes
- D. Cloud cost reductions

QUIZ ANSWER 1

What sparked the DevOps movement?

- A. New hardware advances
- B. Agile adoption challenges**
- C. Regulatory changes
- D. Cloud cost reductions

QUIZ QUESTION 2

Which tool is used for containerization?

- A. Jenkins
- B. Docker
- C. Slack
- D. Prometheus

QUIZ ANSWER 2

Which tool is used for containerization?

- A. Jenkins
- B. Docker**
- C. Slack
- D. Prometheus

QUIZ QUESTION 3

What does IaC improve?

- A. Manual configuration time
- B. Consistency across environments
- C. Team silos
- D. Deployment errors

QUIZ ANSWER 3

What does IaC improve?

- A. Manual configuration time
- B. Consistency across environments**
- C. Team silos
- D. Deployment errors

QUIZ QUESTION 4

What does visibility of work enable?

- A. Reduced collaboration
- B. Better bottleneck identification
- C. Increased delays
- D. Less accountability

QUIZ ANSWER 4

What does visibility of work enable?

- A. Reduced collaboration
- B. Better bottleneck identification**
- C. Increased delays
- D. Less accountability

QUIZ QUESTION 5

What is a DevOps cultural tenet?

- A. Siloed teams
- B. Continuous learning
- C. Blame assignment
- D. Static processes

QUIZ ANSWER 5

What is a DevOps cultural tenet?

- A. Siloed teams
- B. Continuous learning**
- C. Blame assignment
- D. Static processes

QUIZ QUESTION 6

What does deployment frequency indicate?

- A. System downtime
- B. Release agility
- C. Error rates
- D. Manual effort

QUIZ ANSWER 6

What does deployment frequency indicate?

- A. System downtime
- B. Release agility**
- C. Error rates
- D. Manual effort

QUIZ QUESTION 7

Why reduce handoffs in DevOps?

- A. Increase delays
- B. Improve task ownership
- C. Add dependencies
- D. Slow delivery

QUIZ ANSWER 7

Why reduce handoffs in DevOps?

- A. Increase delays
- B. Improve task ownership**
- C. Add dependencies
- D. Slow delivery

QUIZ QUESTION 8

What does the Third Way emphasize?

- A. Static processes
- B. Continuous experimentation
- C. Avoiding failure
- D. Limiting feedback

QUIZ ANSWER 8

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- A. Static processes
- B. Continuous experimentation**
- C. Avoiding failure
- D. Limiting feedback